Mohammad Amin Roshani

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Scholar | LinkedIn | GitHub

Education

Wayne State University 2023 – Present
Ph.D. in Computer Science

Wayne State University 2023 – 2025
M.S. in Computer Science

The University of Memphis 2022 – 2023
Ph.D. studies in Computer Science (transferred to Wayne State) *GPA: 3.94*

The University of Tehran 2017 – 2022
B.Sc. in Computer Science *GPA: 3.1*

National Organization for Development of Exceptional Talents 2013 – 2017
Mathematics High School Diploma *GPA: 4.0*

Interests

Health Informatics, Machine Learning, Natural Language Processing, Data Analysis

Experience

Graduate Student Instructor Jan 2026 – May 2026
Wayne State University *Instructor: Dongxiao Zhu*

- CSC 5991: Trustworthy AI for Large Language Models and Vision-Language Models
- Led weekly labs on LLM/VLM foundations, supervised fine-tuning, prompting, CLIP/LLaVA, red-teaming, unlearning, grounded VLMs (LLaVA+SAM), and agentic AI safety
- Mentored student teams designing, evaluating, and documenting trustworthy AI applications

Graduate Teaching Assistant Aug 2025 – May 2026
Wayne State University

- Problem Solving and Programming – Hadi Nasser

Graduate Teaching Assistant Aug 2024 – May 2025
Wayne State University

- Introduction to Web Technology – Kamel Rushaidat
- Senior Capstone Project – Seyed Ziae Mousavi Mojab

Graduate Research Assistant May 2023 – Aug 2024
Wayne State University *Supervisor: Dongxiao Zhu*

- Working on NIH project about using LLMs to build a conversational app to predict COVID-19 severity risk
- Conducting research on safety of LLMs and multimodal foundational models

Graduate Teaching Assistant Aug 2022 – Apr 2023
The University of Memphis

- CS1: Introduction to Programming (Python) – William Baggett
- Foundations of Computing – Thomas Watson

Private Tutor Feb 2020 – Aug 2022
Self-Employed

- Teaching Introduction to Python to students in Iran and Australia

Teaching Assistant Jan 2020 – Jan 2022
The University of Tehran

- Advanced Information Retrieval (NLP) – Hamidreza Keshavarz
- Fundamentals of Computer Science and Programming – Deniz Tahmouresi
- Machine Language and Assembly – Abbas Nowzari-Dalini
- Principles of Computer Systems – Bagher BabaAli

Publications

Learning to Poison Large Language Models for Downstream Manipulation — Oct 2026
Accepted (short paper), IEEE BigData 2026

Automatic Calibration for Membership Inference Attack on Large Language Models — Jul 2025
Published

Generative Large Language Model-Powered Conversational AI App for Personalized Risk Assessment: Case Study in COVID-19 — Feb 2025
Published

Languages

English (Fluent; TOEFL 106, GRE V 152), Persian (Native), Arabic (Elementary)

Projects

AI-powered web application for elderly health tracking (collaboration with nursing school) — Dec 2025

Adding LLMs to the risk assessment app to make a conversational agent — Mar 2024

Designing COVID-19 risk severity assessment for NIH project — Oct 2023

Hate speech detection using Transfer Learning — Oct 2022

Additional: Sarcasm detection, sentiment analysis, segmentation, Bi-LSTM hate speech, regression/statistics — 2021–2022

Technical Skills

Programming Languages: Python (primary), C++, JavaScript, Go, R, SQL, Bash, HTML, CSS, LaTeX
Machine Learning & AI: Deep Learning, LLMs, Transformers, RAG, LoRA, prompt engineering, multi-agent AI, XAI, clinical NLP
ML Frameworks & Libraries: PyTorch, Hugging Face, Scikit-learn, NumPy, Pandas, TensorFlow, OpenCV, Matplotlib, NLTK
NLP: BERT, GPT-style, LLaMA, tokenization, instruction fine-tuning, information extraction from clinical text
Web & Application Development: React, Node.js, JavaScript, REST APIs, web and mobile-connected apps
Data Science & Modeling: Statistical modeling, regression, classification, experimental design, AUC/F1/precision-recall
Tools & Infrastructure: Git, Linux/Ubuntu, VS Code, GPU pipelines, Hugging Face ecosystem, shell scripting
Research & Writing: LaTeX, reproducible experiments, dataset construction, peer review

Honors

Admitted to the second level of the National Student Olympiad (Mathematics and Literature)
Top 5 of the class for the last four semesters
First place in University Dart Competitions

References

Available upon request.